ASSIGNMENT -4

K. R. MANGALAM UNIVERSITY

B.Tech CSE (AI & ML)

Section - B

SOET (School of Engineering and Technology)

NAME: Nikhil

ROLLNO: 2301730077

Course: Computer Networks
Course Code: (ENCS352)

K. R. MANGALAM UNIVERSITY,

SOHNA ROAD GURUGRAM HARYANA 122102

1. Objective

The objective of this experiment is to configure and test a Domain Name System (DNS) server using Cisco Packet Tracer. The aim is to understand how domain names are resolved into IP addresses and to verify DNS functionality using networking commands.

2. Introduction

The **Domain Name System (DNS)** is a critical component of computer networks that translates human-readable domain names (such as www.google.com) into machine-readable IP addresses (such as 8.8.8.8).

Without DNS, users would need to remember IP addresses instead of domain names.

DNS works using a **query-response mechanism**:

1. Client sends a request (DNS query)
2. DNS server processes the request
3. Server returns the corresponding IP address

3. Network Design

The network topology consists of:

- 1 DNS Server
- 3 PCs (clients)
- 1 Switch

All devices are connected using **copper straight-through cables** within the same network.

IP Configuration

Device	IP Address	DNS Server
Server	192.168.1.2	—
PC0	192.168.1.3	192.168.1.2
PC1	192.168.1.4	192.168.1.2
PC2	192.168.1.5	192.168.1.2

4. DNS Server Configuration

The DNS service was configured on the server using the following steps:

- Enabled DNS service
- Added DNS records:

Domain Name	IP Address
www.example.com	192.168.1.2
www.google.com	8.8.8.8

This allows the DNS server to resolve domain names into corresponding IP addresses.

5. Implementation and Testing

5.1 Ping Test

Command:

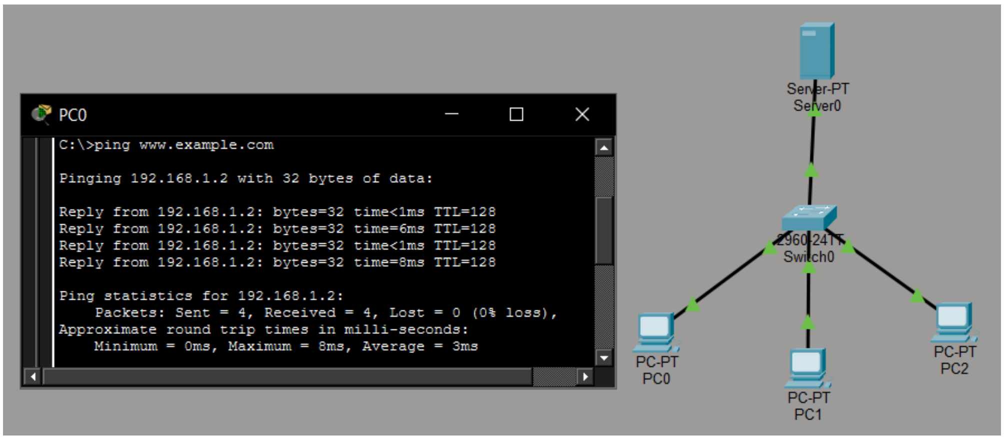
ping www.example.com

Result:

- Successfully resolved to 192.168.1.2
- All packets received without loss

This confirms:

- DNS resolution is working
- Network connectivity is established



5.2 NSLookup Test

Command:

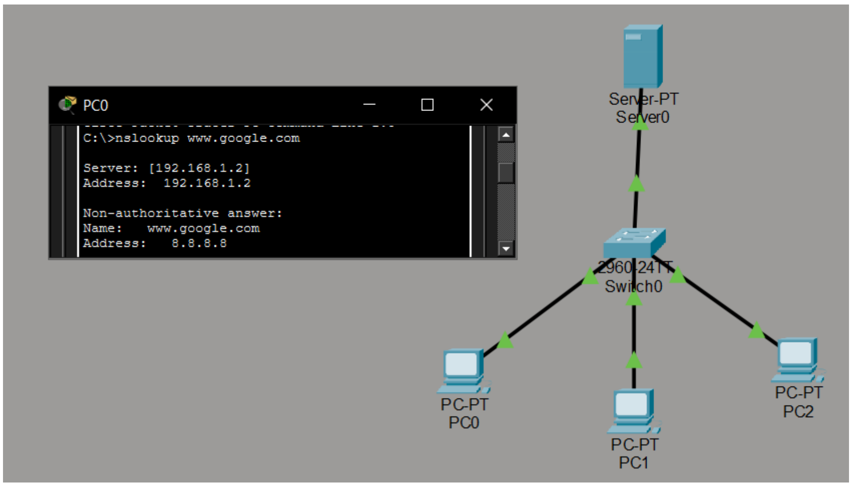
nslookup www.google.com

Result:

- Domain resolved to 8.8.8.8

This confirms:

- DNS server is correctly mapping domain names
- Name resolution process is functional



5.3 Traceroute Test

Command:

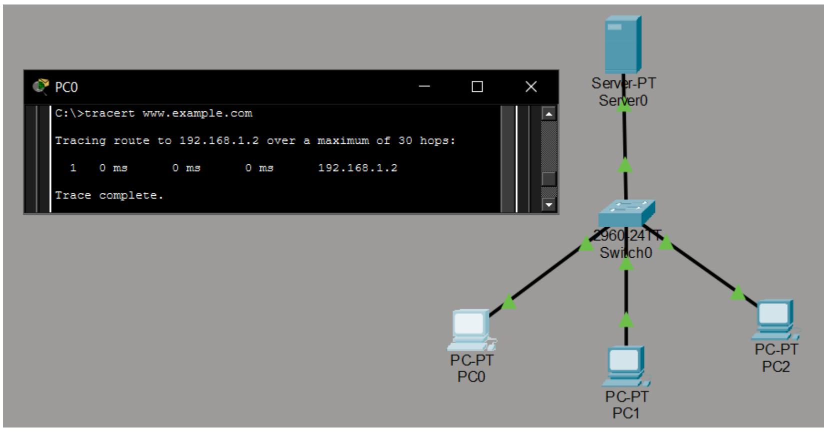
tracert www.example.com

Result:

- Completed in a single hop (192.168.1.2)

Explanation:

- All devices are in the same local network
- No router is involved
- Therefore, the destination is reached directly



6. Analysis

- DNS successfully translates domain names into IP addresses
- Queries are processed locally by the configured DNS server
- The absence of a router limits the network to a single-hop communication
- External addresses like 8.8.8.8 are resolved but not reachable due to lack of internet simulation

7. Conclusion

The experiment successfully demonstrated the configuration and testing of a DNS server in Cisco Packet Tracer.

The DNS server correctly resolved domain names, and connectivity was verified using ping, nslookup, and traceroute commands.

This experiment helped in understanding:

- DNS working mechanism
- Name resolution process
- Basic network configuration